

Exception Flows

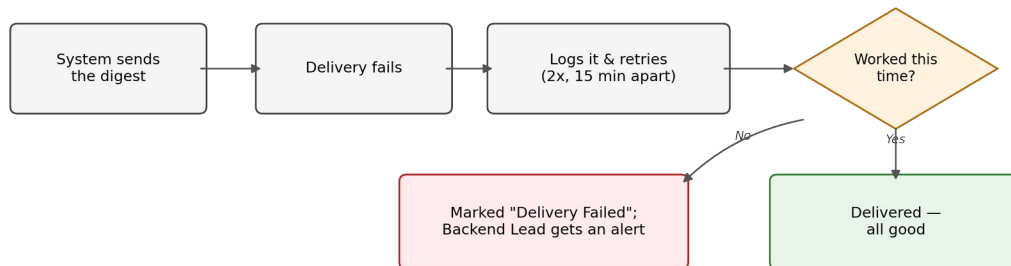
Generate Weekly Optimization Digest (UC-04)

Database Query Performance Profiler — Problem Statement #44

An exception flow kicks in when something actually goes wrong — a service is down, a connection drops, something fails. These describe how the system notices the problem and recovers (or gracefully backs off) instead of just breaking.

EF-01 — Notification Delivery Fails

Sometimes the message just doesn't get through — Slack might be down, or an email bounces. Rather than giving up right away, the system tries a couple more times before raising a flag.



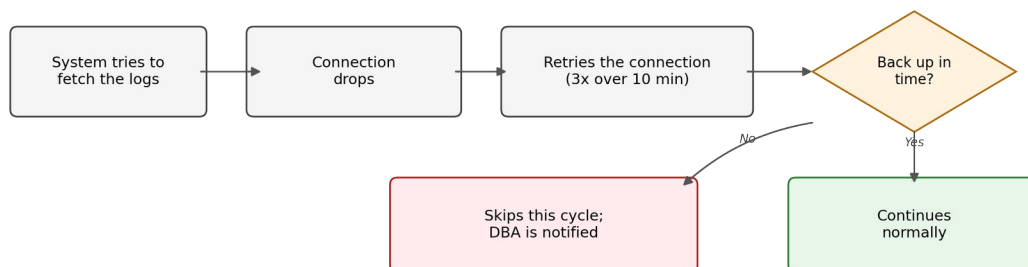
What happens, step by step:

- The system tries to send the digest to the Notification Service.
- The delivery fails.
- It logs what went wrong and retries twice, waiting 15 minutes between each attempt.
- If it's still failing after that, the digest gets marked "Delivery Failed" and the Backend Lead gets an in-app alert instead.
- Nothing is lost — the digest is still sitting on the dashboard, ready to view manually.

Who's involved: Backend Lead, Database Administrator, Notification Service.

EF-02 — Database Connection Lost During Log Retrieval

If the system can't even reach the database to pull the week's logs, there's nothing to build a digest from. Instead of sending something broken or incomplete, it backs off and tries again a little later.



What happens, step by step:

- The system tries to fetch the week's query logs.
- The database connection drops or times out.
- It retries up to 3 times over a 10-minute window.
- If the connection still isn't back, it skips this cycle entirely and lets the DBA know.
- Last week's digest stays available on the dashboard, so nobody's left with nothing.

Who's involved: Database Administrator, Backend Lead.